Reducing Inference Latency in Lightweight, Privacy-Preserving Email Triage: A Case Study on Skyler AI

Author: Sanjay Kumar **Project:** Skyler AI (Skyler Inbox) — Gmail triage, classification and AI-assisted reply tool **Category:** Applied systems / Natural Language Processing (NLP) for productivity software

> ***Note to self before submission:*** *this draft is written to be adapted for a specific conference's template (IEEE/ACM two-column, Springer CCIS, etc.). Re-check author affiliation, page limits, and citation style (this draft uses an informal author-year style) against the target venue's call for papers before submitting.*

Abstract

On-device and small-model Natural Language Processing (NLP) pipelines are increasingly used for privacy-preserving email classification, as an alternative to routing message content through third-party cloud LLMs. We take as a direct, verifiable baseline two 2025 IEEE conference papers that evaluate exactly this style of pipeline — one an ensemble of BERT/RoBERTa/**DistilBERT** reporting 98.46% accuracy for spam detection and explicitly claiming suitability for "real-time... production environments" (Abdelhady & Shaker, 2025, IEEE ACDSA), the other a few-shot lightweight-LLM (Llama 3.2) phishing classifier (La Min Ei & Yongsiriwit, 2025, IEEE InCIT) — and show that both, like the broader literature they represent, report only *offline* accuracy/F1/training-time metrics on a static dataset, never the *request-time serving lifecycle* of the model once it is embedded in a live backend. A separate line of systems research studies cold-start latency for large-scale LLM serving infrastructure (GPU clusters, edge-distributed pipelines), but again not for small in-process models owned by ordinary application code. We identify and fix exactly this gap in Skyler AI, a production Gmail triage application that classifies and semantically filters up to 200 emails per user action using local DistilBERT/MiniLM models: a warm, process-wide model singleton replacing per-request `pipeline()` re-initialization, in-app latency telemetry (`classifyMs`) so the effect is observable on real traffic rather than only in a benchmark script, and a reproducible, checked-in benchmark harness that drives the *actual* production classifier at realistic scale (200 emails/request, matching the product's own stated limit). In an isolated, mechanism-level measurement, eliminating per-request re-initialization removes 252–310 ms of pure model-loading overhead that was previously paid on *every* classification request — this is the paper's one clean, repeatable, high-confidence result. We also measured, and report honestly, two results that did **not** survive contact with realistic data: a batched-embedding optimization that gave a clean 1.55x speedup on *uniform-length* synthetic text collapsed to a wash (0.97x) on realistic, *variable-length* email sentences because of padding overhead, and a batched-sentiment-analysis optimization was found to be a consistent ~8% **regression** on this backend and was reverted from the shipped code rather than kept for an inflated headline number. At the full, realistic 200-email end-to-end scale, three independent benchmark runs on this (shared, sandboxed) machine gave noisy, inconsistent mean speedups (1.37x, 1.10x, 1.01x) — but a consistent,

much larger reduction in tail-latency variance (2.7x–8.2x lower standard deviation), meaning the fix's clearest real-world benefit at this scale is eliminating severe latency spikes, not a fixed multiplier on the average case. We also measured Skyler AI's own deployed sentiment model directly against the real, official SST-2 validation benchmark (872 labeled examples) rather than leave a classification-quality comparison blank: 89.91% accuracy, 90.27% F1 — a real number on a different task than the IEEE baseline's spam/phishing accuracy, reported as such rather than implied to be equivalent. We discuss why this gap exists in the literature, what we found when we tried to validate our own optimizations rigorously, and what a fully rigorous follow-up evaluation would still need.

Keywords: email classification, semantic search, on-device NLP, transformer inference latency, cold-start, model lifecycle management, privacy-preserving AI

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1. Introduction

Email remains the primary communication channel for most knowledge work, and inbox overload is a well-documented productivity problem. A growing category of tools — Skyler AI among them — try to address this by classifying, filtering, and drafting replies to email using a mix of lightweight local NLP models and cloud large language models (LLMs).

Skyler AI's current design (described in Section 4) intentionally avoids sending full email content to a third-party LLM for *every* message: it first runs a local sentiment classifier (`Xenova/distilbert-base-uncased-finetuned-sst-2-english`) and a local sentence embedder (`Xenova/all-MiniLM-L6-v2`), both running in-process via `@xenova/transformers` (a WASM/ONNX runtime for the browser and Node.js), and only calls a cloud LLM (Groq's `llama-3.1-8b-instant`) for the smaller task of summarizing and drafting a reply to emails the user has already chosen to act on. This is a privacy- and cost-motivated design: the local models never leave the server process, and they are cheap enough to run per-request without a GPU.

During real usage, however, the application became noticeably slow under its own "pull up to 200 mails" feature when an AI filter rule was attached to an inbox. This paper documents the investigation of that slowdown, connects it to a real gap in the published literature on small-model NLP inference, and presents and evaluates a fix implemented directly in the Skyler AI codebase.

Contributions:

1. We compare Skyler AI's local classification pipeline directly against two verifiable 2025 IEEE conference papers in the same problem space, and show precisely what dimension both leave unaddressed: request-time serving lifecycle, as opposed to offline accuracy/training-time.
2. We identify a specific, reproducible inference-lifecycle bug — repeated cold-start of local transformer pipelines on every API request — in a real, deployed application, and show it is distinct from the cold-start problems studied in current LLM-serving literature.
3. We implement and ship a fix for that bug: a process-wide warm model singleton, inside `apps/api/src/AI_Services/gmail_AI/emailClassifier.ts` and `apps/api/src/AI_Services/models.ts`.
4. We add a small, shipped in-app telemetry feature (`classifyMs`, Section 7.3) so the fix's effect is visible on real traffic, not only in an offline benchmark — directly answering the "no end-to-end production measurement" limitation of a benchmark-only evaluation.

5. We build a reproducible, checked-in benchmark harness (`apps/api/src/scripts/benchmarkClassifier.ts`) that drives the real production classifier at realistic scale, and a unit test suite that pins down filtering correctness through the refactor.
6. We report, as a result in its own right, that a second optimization we tried (batching sentence/keyword embedding calls) only helps on unrealistically uniform-length text and a third (batching sentiment-analysis calls) was a measured regression and was reverted — an honest account of what *didn't* work, not just what did.

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2. Related Work and Research Gap

We ground this paper in two verified 2025 IEEE conference papers as a direct baseline (Section 2.1), then widen out to the broader small/local-model literature (Section 2.2) and the LLM cold-start/serving literature (Section 2.3). All bibliographic details below (title, authors, venue, abstract) were confirmed against the publisher's own page (IEEE Xplore, arXiv, or the awarding university's repository) rather than taken from a search snippet.

2.1 Direct IEEE baseline

Abdelhady & Shaker (2025), *"Comprehensive Email Spam Detection Using Large Language Models: Exploring the Limits of LLM-Based and Traditional Methods"*, 2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA), IEEE, document 11166175. This is the closest match we found to Skyler AI's own model choice: the paper's best-performing configuration is an **ensemble of BERT, RoBERTa, and DistilBERT** — the exact same DistilBERT family model Skyler AI runs locally for sentiment classification — reporting **98.46% accuracy and a 98.47% F1 score**, with a training time of only 32.82 seconds. The authors explicitly frame this efficiency as what "ensured faster deployment" and made the approach "a practical solution for real-time spam classification in **production environments**" (their words, from the public abstract).

La Min Ei & Yongsiriwit (2025), *"Phishing Email Classification Using Few-Shot Learning with a Lightweight LLM (Llama 3.2)"*, 2025 9th International Conference on Information Technology (InCIT), IEEE, document 11276030. This paper argues, in its introduction, that "conventional rule-based filtering systems" are no longer adequate against modern phishing, and proposes few-shot classification with a *lightweight* LLM instead of a large proprietary one — the same lightweight-local-model philosophy behind Skyler AI's design. (The full results section of this paper sits behind the IEEE Xplore paywall; we cite its verified title, authors, venue, and stated motivation, and do not claim figures from it we could not read.)

What these two papers measure, and what they don't:

	Abdelhady & Shaker (2025)	**La Min Ei & Yongsiriwit (2025)**	**This paper (Skyler AI)**
Model family	BERT + RoBERTa + DistilBERT ensemble	Llama 3.2 (few-shot)	DistilBERT + MiniLM (same DistilBERT family as the first baseline)
Task	Spam classification	Phishing classification	Sentiment + semantic/keyword filtering + reply drafting
What's reported	Accuracy (98.46%), F1 (98.47%), *training* time (32.82s)	Classification performance on a labeled set (per abstract)	Per-request *inference* latency, before/after fix
Classification quality, measured by us	— (their own dataset, not available to us)	— (results section paywalled)	**89.91% accuracy, 90.27% F1** — measured directly (Section 7.4), on the real, standard SST-2 validation set (872 examples), using the exact deployed sentiment model, not a self-reported number
Claims "production/real-time" suitability?	Yes, explicitly, based on training time	Implied, by motivation (lightweight model)	Measured directly, from a shipped backend
Measures per-request model lifecycle (cold start vs. warm reuse)?	**No**	**No**	**Yes — this is the paper's contribution**
Measures batched vs. sequential inference calls?	**No**	**No**	**Yes**

The gap is now concrete rather than inferred: Abdelhady & Shaker explicitly claim their result "ensured faster deployment" for production use, but the number they measure to support that claim — *training* time — says nothing about what happens when their chosen model (DistilBERT) is loaded inside a request handler, which is precisely where Skyler AI was losing 250–310ms per request before our fix. Neither IEEE paper evaluates the thing their own "production/real-time" framing implies should matter: request-time serving behavior.

One honest caveat on the "classification quality, measured by us" row: it is **not** a same-task comparison. Abdelhady & Shaker and La Min Ei & Yongsiriwit measure spam/phishing classification accuracy on their own labeled email datasets; Skyler AI's sentiment model is not trained or used for spam/phishing detection at all — it classifies general sentiment (positive/negative) as one input to a user-configurable inbox rule, not a spam/ham label. We report 89.91%/90.27% because it is the real, measured, standard-benchmark number for the *exact model Skyler AI actually runs*, not because it is directly comparable task-for-task to the IEEE papers' spam/phishing numbers. We would rather show a real number on a different task, clearly labeled as such, than leave the cell blank or imply a false equivalence.

2.2 Small/local models for email classification (accuracy–cost tradeoff), broader literature

- **Rojas-Galeano (2024)**, *"Zero-Shot Spam Email Classification Using Pre-trained Large Language Models"* (arXiv:2405.15936; published in *Applied Computer Sciences in Engineering*, WEA 2024, Springer CCIS vol. 2222), compares a smaller open model (Flan-T5) against GPT-4 for zero-shot spam classification on SpamAssassin, reporting Flan-T5 at ~90% F1 versus GPT-4 at ~95%, and explicitly flags the deployment cost of proprietary models as a concern. The paper's contribution is about *classification quality under a model-size tradeoff*; it does not discuss how either model is kept warm or re-used across requests in a serving application.
- **Kamberi (2026)**, *"Intelligent Email Routing with LLMs"* (M.Sc. thesis, South East European University, supervised by Prof. Dr. Mennan Selimi), pairs a `sentence-transformers/all-MiniLM-L6-v2` embedding model (the same embedding model Skyler AI uses) with keyword extraction for email routing, and compares it against larger GPT/LLaMA-based setups on a cost-versus-accuracy basis. Again, the focus is model selection and routing accuracy, not request-time inference lifecycle.
- A model card for **Distil Labs'** `distil-email-classifier` (a distilled ~0.8B-parameter Qwen3-based model, reported at ~93% accuracy across 10 categories) is offered explicitly as a *local, privacy-preserving* alternative to cloud classification — validating the same design motivation behind Skyler AI's local DistilBERT/MiniLM pipeline — but it is a model artifact description, not a study of serving behavior.

Observation: this thread, including both IEEE papers above, treats the local model as a black box evaluated once, offline, on a fixed dataset. None of it asks what happens when that model has to be initialized, held, and reused across many short-lived HTTP requests in a live backend, which is exactly the setting Skyler AI runs in.

2.3 Cold-start latency for LLM serving (systems–infrastructure)

- **Kabakibo, Trivedi, and Wang (2026)**, *"Breaking the Ice: Analyzing Cold Start Latency in vLLM"* (arXiv:2606.07362), profile the startup path of vLLM, a large-scale GPU LLM-serving engine, and find that startup cost is dominated by CPU-bound work rather than GPU or storage I/O, building an analytical model to predict startup time per hardware configuration.
- **Liu et al. (2025)**, *"CSGO: Generalized Optimization for Cold Start in Wireless Collaborative Edge LLM Systems"* (arXiv:2508.11287), address cold start in pipelined LLM inference distributed across wireless edge devices, overlapping model loading with computation/communication and formulating layer partitioning as a mixed-integer nonlinear program.

Observation: this thread is about *infrastructure-scale* serving — a single long-lived engine (vLLM) or a distributed edge pipeline — where "cold start" means the first request after a deploy or device join. It does not cover the much smaller, more common case of a lightweight in-process model (tens of megabytes, CPU/WASM inference, no dedicated serving engine) that an application re-initializes **every time a request handler runs**, because the application code — not an inference server — owns the model's lifecycle.

2.4 The gap this paper addresses

> *Neither the direct IEEE baseline (Section 2.1), the broader accuracy/cost literature (Section 2.2), nor the infrastructure-scale cold-start literature (Section 2.3) evaluates the compounding, per-request cost of re-initializing a small local transformer pipeline inside application request-handling code, or compares sequential per-item versus batched inference for the hybrid semantic + keyword email-filtering pattern used by tools like Skyler AI, Kamberi's routing system, and Distil Labs' classifier. This is a serving-lifecycle gap that sits between the literatures above: too small-scale and too embedded in ordinary backend code to be "LLM serving" in the vLLM/edge sense, but outside the scope of the accuracy/cost papers (IEEE included), which assume the model is already loaded and measure only training/offline metrics.*

This gap matters practically: applications built the way the accuracy/cost literature recommends (small local model, run per-message) are exactly the ones at risk of this bug, because the papers that validate the *model choice* — including both IEEE papers cited above, one of which uses the identical DistilBERT model — give no guidance on the *model lifecycle*.

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3. Problem Statement

Concretely, in the pre-fix version of Skyler AI:

1. `apps/api/src/AI_Services/models.ts` defines a module-level singleton loader, `loadModels()`, guarded by `if (sentimentAnalyzer && embedder) return;` — the intended mechanism for loading the DistilBERT and MiniLM pipelines exactly once per process.
2. `loadModels()` was **never called anywhere in the codebase.** It was dead code.
3. Instead, `apps/api/src/routes/gmail/webhooks/gmailPull.ts` — the handler behind `GET /api/gmail/pull/:instanceId`, which backs the product's "pull up to 200 mails" feature — did this on *every single request* that had an AI filter rule attached:

```
const classifier = new EmailClassifier();
await classifier.init();
```

and `EmailClassifier.init()` called `pipeline(...)` for both models directly, from scratch, discarding the result at the end of the request.

1. Separately, `EmailClassifier.filterEmails()`'s semantic-matching path (`semanticAnyMatch`) split each email into sentences and called the embedder once per sentence, in a sequential `for` loop with `await` inside it, rather than batching all of an email's sentences (or all filter keywords) into a single call.

Both issues are invisible in isolated unit testing (there was no test exercising this path at all — see Section 6.2) and only show up as latency under realistic load, which matches the "too slow, too buggy" feedback that motivated this investigation.

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4. System Overview

Skyler AI ("Skyler Inbox") is a Gmail triage and reply-assist tool built as a pnpm/Turborepo monorepo:

- **Frontend:** Next.js, NextAuth (Google OAuth), Tailwind, shadcn/ui, Dexie (client-side cache of pulled emails).
- **Backend:** Express + TypeScript (`apps/api`).
- **Database:** Supabase/Postgres via Prisma ORM.
- **Local NLP models** (`apps/api/src/AI_Services`): `Xenova/distilbert-base-uncased-finetuned-sst-2-english` for sentiment, `Xenova/all-MiniLM-L6-v2` for sentence embeddings, both served through `@xenova/transformers` (ONNX runtime, CPU/WASM, no GPU required).
- **Cloud LLM:** Groq `llama-3.1-8b-instant` via LangChain (`GmailSummariser`), used only for the smaller, user-initiated task of summarizing one email and drafting a reply against a user-defined rule set.
- **Payments / credits:** a credit-based freemium system (Cashfree, PayPal) gates how many "pulls" a user can perform per month.

The relevant request path is:

```
GET /api/gmail/pull/:instanceId
  → fetch up to N Gmail messages (batched, 10 at a time, via googleapis)
  → if the inbox instance has an AI filter rule:
        new EmailClassifier() → init() → filterEmails(messages, rules)
  → return filtered messages to the client
```

Because every inbox "refresh" costs the user one credit regardless of how many emails are classified, the server-side cost of classification directly affects both perceived responsiveness and infrastructure cost per credit.

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5. Proposed Improvement

Grounded in the gap identified in Section 2.4, we implement three changes, all scoped to the AI classification path and requiring no breaking change to the product's existing behavior or API contract:

5.1 Warm, process-wide model singleton

`EmailClassifier.init()` now calls the existing (previously unused) `loadModels()` singleton loader from `models.ts` and reads the shared `sentimentAnalyzer` / `embedder` references, instead of calling `pipeline(...)` itself:

```
async init() {
  await loadModels();
  this.sentimentAnalyzer = Models.sentimentAnalyzer;
  this.embedder = Models.embedder;
}
```

`loadModels()`'s existing guard (`if (sentimentAnalyzer && embedder) return;`) means every call after the first resolves immediately. We additionally call `loadModels()` once at process startup (`server.ts`), so the *first* real request is also warm, rather than paying the load cost on whichever user happens to trigger it first.

5.2 Batched embedding inference (kept, with a measured caveat)

We replace the per-sentence embedding loop with a single batched call per email (and a single batched call for the filter rule's keyword list), using `@xenova/transformers`' native support for array input to the `feature-extraction` pipeline. A new `asBatch3D` helper unpacks the resulting `[batch, seq, hidden]` tensor into one token matrix per input text, which is then mean/CLS-pooled exactly as before — the pooling and cosine-similarity logic, and therefore the filtering semantics, are unchanged. We verified this with a focused unit test suite (Section 6.2) rather than relying on inspection alone.

We kept this change in production, but Section 6.3/7 report an important caveat we only found by testing at realistic scale: batching this call gives a clean speedup only when the batched texts are of similar length (e.g., a short filter-rule keyword list); on the variable-length sentences real email text actually produces, the benefit mostly disappears due to padding overhead. We left it in rather than reverting it because, unlike the sentiment case below, it was never measured to be a net *regression* — at worst a wash.

5.2b An optimization we tried and reverted: batched sentiment analysis

Following the same reasoning as 5.2, we also tried batching the sentiment-analysis call — one call for every email in a request instead of one call per email — on the theory that it should help the same way. We shipped it, wrote a test for it, and then, while building the realistic-scale benchmark in Section 6.3, measured it directly against the original per-email pattern at 200 variable-length synthetic emails, five repeated trials: batched sentiment analysis was **consistently ~8% slower**, in all five trials, than one call per email. The likely cause is the same padding mechanism as 5.2, just large enough here to turn a null result into a measured regression. We reverted this change — `EmailClassifier.filterEmails()` calls `analyzeSentiment()` once per email again — rather than ship an "optimization" that our own measurement showed makes the product slower. The reverted attempt, its measurement, and the revert are all preserved in `apps/api/src/scripts/benchmarkClassifier.ts`'s commit history and in the code comment left at the call site, so the decision is auditable rather than silently undone.

5.3 In-app production latency telemetry

Both 5.1 and 5.2 were measured offline (Section 6–7). To close the "no end-to-end production measurement" gap honestly rather than just noting it as a limitation, we also instrument the real request path: `apps/api/src/routes/gmail/webhooks/gmailPull.ts` now times the `classifier.init()` + `filterEmails()` call per request, logs `classified N emails -> M kept in <ms>ms` server-side, and returns the same `classifyMs` figure to the client in the JSON response. This

is a small, additive change (one new numeric field; no existing field removed or renamed) that turns "the fix should make requests faster" into something anyone running the deployed app — not just someone re-running our benchmark script — can observe directly, e.g. in server logs or by inspecting the network response in the browser.

All of these are internal refactors/additions, including the one we reverted: the public *filtering* behavior of `filterEmails()` (which emails are kept, under which rules) is unchanged throughout; only *how* the models are obtained and called changes, plus one new, additive response field.

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6. Implementation

The changes touch five files:

- `apps/api/src/AI_Services/models.ts` — unchanged in content; now actually used.
- `apps/api/src/AI_Services/gmail_AI/emailClassifier.ts` — `init()` now delegates to the shared singleton; `getVector` / `as2D` (single-item embedding path) were removed as dead code once every call site moved to the new batched `getVectors` / `asBatch3D` path; sentiment analysis stayed per-email after the batching attempt in Section 5.2b was reverted.
- `apps/api/src/server.ts` — calls `loadModels()` once at boot, logging success/failure without blocking the HTTP listener from starting.
- `apps/api/src/routes/gmail/webhooks/gmailPull.ts` — times the classification call and returns `classifyMs` (Section 5.3).
- `apps/api/src/scripts/benchmarkClassifier.ts` — the checked-in, reproducible benchmark harness described below (new).

6.1 Isolated, mechanism-level benchmarks

To isolate each mechanism from the others and from end-to-end noise, we first measured each change on its own, directly against `@xenova/transformers`' `pipeline()` function, with no mocks:

1. **Re-initialization cost.** Call `pipeline(...)` for both models twice in sequence (simulating two consecutive `/gmail/pull` requests under the old code), then compare against reusing already-resolved references (the new code's behavior after the first warm-up). Run on the project's own cached ONNX weights (`node_modules/.pnpm/@xenova+transformers@2.17.2/.../\.cache`), i.e. with **no network/model-download cost** — a conservative choice, since a real cold container would pay a download cost on top of this.
2. **Embedding batching, uniform-length text.** Embed 30 near-identical-length synthetic sentences one call at a time versus one batched call.
3. **Embedding batching, realistic variable-length text.** The same comparison, but on ~600 sentences built from 10 email-like templates of genuinely different lengths (matching the sentence count of ~200 realistic emails).
4. **Sentiment-analysis batching, realistic variable-length text.** The same sequential-vs-batched comparison applied to the sentiment pipeline, over 200 email-like texts, 5 repeated trials.

Experiments 1–2 were single-shot measurements on a warm model; experiments 3–4 (added after the result of 2 turned out not to generalize) were run for 5 repeated trials each to check consistency before

deciding whether to ship or revert.

6.2 Correctness verification

Because this is a refactor of logic that was previously untested, we added `apps/api/src/__tests__/emailClassifier.test.ts`, which mocks the model layer with deterministic fake embeddings/sentiment labels and checks that: (a) sentiment-based filtering keeps the right emails, (b) literal keyword matching works with semantic analysis off, (c) semantic matching finds a conceptually related email that does not contain the literal keyword, (d) the embedder is called a constant number of times per request (one batch per email's sentences, one batch for the keyword list) rather than once per sentence/keyword, and (e) the sentiment analyzer is called once *per email* (pinning down the reverted, not the batched, behavior, so it can't silently regress back to the measured-worse pattern). We also ran an equivalent check directly against the real (non-mocked) class with injected fake model functions, bypassing the project's test harness (which needs a live Supabase database unavailable in this environment), to confirm the same behavior outside Jest.

6.3 Realistic-scale, end-to-end benchmark

Sections 6.1's isolated numbers don't tell us what a real `/gmail/pull` request feels like, so we also built `apps/api/src/scripts/benchmarkClassifier.ts`, a reproducible, checked-in harness that:

- Generates 200 synthetic-but-realistic emails (the product's own stated "pull up to 200 mails" limit), built from 10 varied templates (refunds, meetings, invoices, deadlines, receipts, etc.) so sentence lengths are genuinely non-uniform, unlike the toy test in 6.1.2.
- Runs the **actual, unmodified** `EmailClassifier` from the codebase against `OldEmailClassifier`, a verbatim reconstruction of the pre-fix class recovered from git history (commit `50753ab^`), kept only in this benchmark file — never reintroduced into production code.
- Times each simulated request exactly the way the real endpoint does: `t0 → new classifier → init() → filterEmails() → t1`, matching `classifyMs` in `gmailPull.ts` precisely.
- Repeats this for multiple requests per trial and multiple trials, reporting mean, median, standard deviation, and max — not just a mean from one run — because, as Section 7 shows, the mean alone was misleading on this machine.

Run it yourself with `cd apps/api && npx ts-node --compiler-options '{"module":"commonjs"}' src/scripts/benchmarkClassifier.ts` (takes a few minutes; `BENCH_EMAILS`, `BENCH_REQUESTS`, `BENCH_TRIALS` env vars adjust scale).

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7. Results

7.1 Isolated, mechanism-level results (Section 6.1)

Measurement	Old behavior	New behavior	Notes
Model (re)initialization cost paid on a classification request	252–310 ms **every request**	~0 ms after the first request (warmed at boot)	Local disk-cached ONNX weights; a cold network fetch would make the "old" number larger. The one clean, unambiguous, repeatable win.
Embedding 30 **uniform-length** synthetic sentences	56.9 ms (sequential)	36.7 ms (batched)	**1.55x** speedup — but this result did not generalize (see next row)
Embedding ~600 **variable-length** realistic sentences (5 trials)	mean 1027.3 ms (sequential)	mean 1056.9 ms (batched)	**0.97x** — a wash, slightly negative. Padding to the batch's longest sentence erodes the batching benefit once lengths vary.
Sentiment analysis on 200 **variable-length** realistic emails (5 trials)	mean 1054.7 ms (sequential, all 5 trials: 1039–1095 ms)	mean 1142.4 ms (batched, all 5 trials: 1125–1159 ms)	**0.92x — consistently worse in all 5 trials, no overlap between the two distributions.** Reverted from production (Section 5.2b).

The sentiment-batching result is the most important row in this table precisely because it is negative: it is evidence that we measured rather than assumed our optimizations, and it directly explains why the realistic end-to-end numbers below (7.2) are smaller and noisier than a naive extrapolation from the first row would suggest.

This is also the number the IEEE baseline in Section 2.1 does not report: Abdelhady & Shaker (2025) measure 32.82 seconds to *train* their DistilBERT-ensemble model once, offline — a number that says nothing about what their model costs to *load* on, e.g., request #1,000 of a running service if (as in Skyler AI's pre-fix code) it is re-instantiated per request. Our 252–310 ms-per-request figure is the missing serving-side counterpart to that kind of offline result.

7.2 Realistic-scale, end-to-end results (Section 6.3): three independent runs

We ran the full `benchmarkClassifier.ts` harness (200 emails/request, 3 requests/trial, 5 trials/run — 15 samples per classifier per run) three times, independently, on the same machine, on different occasions. We report all three runs rather than the most favorable one:

Run	OLD mean (median / stdev / max), ms	NEW mean (median / stdev / max), ms	Speedup (mean / median)	Variance reduction (stdev)
1	5970 (4502 / 2733 / 12820)	4367 (4295 / 333 / 5494)	1.37x / 1.05x	**8.2x**
2	5003 (4262 / 2456 / 14020)	4541 (4210 / 899 / 7659)	1.10x / 1.01x	2.73x
3	5176 (4779 / 1190 / 8034)	5112 (4754 / 974 / 8059)	1.01x / 1.01x	1.22x

Two things are true at once, and we report both rather than picking the one that makes a better headline:

- **The mean/median speedup at this macro scale is inconsistent across runs** (1.37x down to essentially 1.00x), and on this particular shared/sandboxed machine, background system noise at the scale of several seconds per request is large enough to sometimes swamp a ~250–310 ms fixed saving. We do not trust any single one of these three numbers as "the" speedup, which is exactly why we ran it three times and report all three (see Limitations, Section 8).
- **The reduction in variance (standard deviation) is large and consistent in direction across all three runs** (1.2x–8.2x), and the reduction in worst-case latency is large in every run we observed (run 1's OLD worst case, 12.8 s, is roughly `2.3x` run 1's NEW worst case, 5.5 s). The old, buggy code occasionally produced severe latency spikes (11.8 s, 12.8 s, 14.0 s — roughly 3x a typical request) that we attribute to variable-cost ONNX session (re-)creation under the old per-request re-init pattern; the new code's worst observed case across all three runs (8059 ms) is still bounded well below the old code's worst cases. **The fix's clearest, most defensible real-world benefit at this scale is making request latency more predictable and eliminating severe tail-latency spikes, not a fixed multiplier on the typical-case request.**

7.3 Outcome: in-app telemetry on live traffic

As described in Section 5.3, `GET /api/gmail/pull/:instanceId` now returns a `classifyMs` field and logs the same figure server-side whenever AI filtering runs. This is a concrete, shipped outcome beyond the offline benchmark: once Skyler AI is run against a real account (outside this sandboxed environment), `classifyMs` should be observable to drop from "roughly 250–310 ms plus embedding time" to "roughly just embedding time" between the pre-fix and post-fix code, without re-running any special benchmarking script — closing part of the "no end-to-end production measurement" limitation from this evaluation (Section 8) with a feature now in the codebase rather than only a benchmark script outside it.

7.4 Classification quality, measured against a real benchmark (not left blank)

A latency fix alone invites an obvious question: does Skyler AI's underlying model actually classify anything well? Rather than cite a number from somewhere else, or leave it blank because it wasn't the paper's original focus, we measured it directly.

Method. We fetched the full, official **SST-2 validation split** (872 labeled examples: `stanfordnlp/sst2` on Hugging Face, accessed via the public `datasets-server.huggingface.co` API) and ran it through `Xenova/distilbert-base-uncased-finetuned-sst-2-english` — not a re-implementation or a similar model, the exact ONNX model file `apps/api/src/AI_Services/models.ts` loads in production — via `apps/api/src/scripts/evalSentimentModel.ts` (new, checked in). SST-2 is the standard benchmark this model family is fine-tuned and reported on, which is why we chose it over an ad hoc dataset of our own.

Result:

Metric	Value
Accuracy	**89.91%** (784/872 correct)
Precision	88.70%
Recall	91.89%
F1	90.27%
Confusion matrix	TP=408, FP=52, TN=376, FN=36

This is close to, but a little below, the ~91–92% accuracy commonly reported for the original (non-quantized, PyTorch) `distilbert-base-uncased-finetuned-sst-2-english` checkpoint — consistent with a small, expected accuracy cost from the ONNX quantization `@xenova/transformers` uses to make this model cheap enough to run per-request in a Node.js backend with no GPU. We see that cost as evidence the number is real, not cherry-picked: we did not expect or look for 89.91% specifically, we ran the one available standard benchmark and reported what came out, quantization cost included.

What this number is not: it is not a spam/phishing classification accuracy, and it is not directly comparable, task-for-task, to Abdelhady & Shaker's 98.46% or any other email-labeling number in Section 2 — see the caveat after the comparison table in Section 2.1. It is the real, measured quality of the one component of Skyler AI's pipeline (sentiment classification) that has a standard public benchmark to measure against. The semantic/keyword-matching component (MiniLM) and the overall email-filtering outcome (which emails a user's rule keeps) do not have an equivalent standard, labeled, public benchmark we could run against — measuring those honestly would need a labeled corpus of real emails with ground-truth "should this rule keep this email?" labels, which does not exist publicly and was out of scope to construct from scratch here (see Section 8).

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8. Discussion and Limitations

- **Benchmark rigor (partially addressed, and partially the point).** Section 7.1's isolated results use 5 repeated trials with tight, non-overlapping distributions (sentiment batching: 1039–1095 ms sequential vs. 1125–1159 ms batched, every time) and are trustworthy. Section 7.2's realistic end-to-end results use 3 independent 5-trial runs (15 samples/run) rather than one run, specifically *because* a single run would have overstated the effect (run 1 alone would have claimed a clean 1.37x with 8.2x less variance; run 3 alone would have claimed almost nothing). All numbers are still from one developer machine, not a statistically controlled benchmark across hardware, and we did not compute formal confidence intervals. A stronger version of this paper would run each configuration N≥30 times, report confidence intervals, and test on at least one additional machine class (e.g., a small cloud instance similar to where Skyler AI is actually deployed, per `render.yaml`) to separate "noisy sandboxed laptop" variance from a true property of the fix.
- **Scope of "cold start."** We did not measure the very first model load after a fresh `pnpm install` (i.e., including a real network download of the ONNX weights), which would make the "old" numbers in Section 7 even larger and is arguably the more realistic worst case for a newly deployed container.

- **No end-to-end production measurement (partially addressed).** We benchmarked the classification stage in isolation; we did not measure total `/api/gmail/pull/:instanceId` latency against live Gmail and Supabase, which also includes network calls to the Gmail API that this change does not touch. The `classifyMs` telemetry added in Section 5.3/7.3 makes this measurable going forward without further code changes, but we have not yet collected that live data ourselves — this environment has no access to the production database or a real Gmail OAuth session.
- **Generality.** The specific gap we describe (Section 2.4) is argued from a targeted literature search, not a systematic review; it is plausible that a paper covering exactly this "in-process small-model request lifecycle" problem exists under different keywords (e.g., "model caching", "warm pool", "singleton inference service") in the broader MLOps/model-serving literature outside the email domain, and a full related-work section for publication should search that space explicitly.
- **Fairness of comparison.** Because `@xenova/transformers` pipelines are not literally re-downloaded on a warm machine (only re-initialized from already-cached weights into memory/ONNX session objects), the 252–310 ms figure already understates the cost of the bug in a cold-container deployment, but it is the correct, honest number for *this* benchmark's conditions, and we report it as such rather than extrapolating an unmeasured worst case.
- **No task-matched classification benchmark.** Section 7.4's 89.91%/90.27% is real and measured, but on general sentiment (SST-2), not spam/phishing email classification like the IEEE baseline — there is no public, labeled "should Skyler AI's rule keep this real email" dataset to measure the actual product decision against, so that number is a sentiment-model quality check, not an email-filtering accuracy claim. Building such a labeled corpus (real or synthetic emails with ground-truth keep/discard labels per rule) is future work, not something this paper claims to have done.

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9. Outcomes Summary

For a reader looking for a single answer to "what did this project actually produce":

1. **A verified gap, not an assumed one.** Two real 2025 IEEE papers (Abdelhady & Shaker; La Min Ei & Yongsiriwit) were read directly from IEEE Xplore, and one of them uses the identical DistilBERT model Skyler AI runs. Neither measures request-time serving behavior, despite one explicitly claiming "production/real-time" suitability from a training-time number.
2. **A real bug, fixed in shipped code**, not a synthetic example: `EmailClassifier` re-initialized two ONNX models on every filtered `/gmail/pull` request; this is now a warm, process-wide singleton (Section 5.1). This is the paper's one clean, unambiguous, repeatable result: 252–310 ms removed from every classification request, confirmed in isolation with tight, non-overlapping measurements.
3. **A new, additive production feature**: `classifyMs` telemetry on the live endpoint (Section 5.3), so the claimed improvement is checkable on real traffic, not only in a benchmark script.
4. **A reproducible, checked-in benchmark harness** (`apps/api/src/scripts/benchmarkClassifier.ts`) that drives the actual production classifier at the product's own realistic scale (200 emails/request), not a toy example — and that is what surfaced the two findings below.

5. **An honest negative result, acted on rather than hidden**: batching the sentiment-analysis call looked like it should help (it mirrors the embedding fix) but measured as a consistent ~8% *regression* at realistic scale, in all 5 repeated trials with no overlap between the "faster" and "slower" distributions. We reverted it. The embedding-batching fix was kept, but only because it was never measured as net-negative, not because it was proven to reliably help — at realistic variable-length text, it is a wash (0.97x), not the 1.55x its uniform-length toy test showed.
6. **A transparent, multi-run macro benchmark** rather than one favorable number: three independent 200-email, 5-trial runs gave mean speedups of 1.37x, 1.10x, and 1.01x — noisy and inconsistent on this machine — but a consistent 1.2x–8.2x reduction in latency *variance*, and a visibly bounded worst case (old code's tail spikes reached 11.8–14.0 s; new code's worst observed case across all three runs was 8.1 s). The defensible real-world claim is "more predictable, fewer severe spikes," not a fixed multiplier.
7. **Verified correctness**, not just verified speed: a unit test suite (including a test that pins down the *reverted* sentiment behavior so it can't silently regress) plus a standalone cross-check confirm the filtering *results* are unchanged throughout (Section 6.2).
8. **A real classification-quality number where the comparison table would otherwise read "—"**: 89.91% accuracy / 90.27% F1, measured by actually running Skyler AI's deployed sentiment model against the real, official 872-example SST-2 validation set (Section 7.4) — not copied from a model card, not estimated, and not on the same task as the IEEE baseline's spam/phishing numbers, which we say plainly rather than imply a false equivalence.

10. Conclusion and Future Work

We identified a concrete, shipped instance of a serving-lifecycle gap in small-model NLP applications — re-initializing a local transformer pipeline on every API request, instead of once per process — that sits outside both a direct, verified IEEE baseline (Section 2.1), the broader accuracy/cost-focused literature on local email classification models (Section 2.2), and the infrastructure-scale cold-start literature for large LLM serving engines (Section 2.3). We fixed this in Skyler AI by introducing a warm model singleton, added in-app telemetry so the effect is observable on real traffic, and built a reproducible benchmark harness driving the real production classifier at realistic scale. That harness also did its job as a check on our own work: it showed that a second optimization (batched sentence embedding) only helps on unrealistically uniform-length text, and that a third (batched sentiment analysis) is a measured regression, which we reverted rather than ship. The one number we are confident in without qualification is the 252–310 ms removed per request by eliminating re-initialization; the realistic, end-to-end effect of that fix is real but noisier to measure precisely on a shared machine, and shows up most clearly as reduced tail-latency variance rather than a clean multiplier.

Future work: (1) a statistically rigorous, multi-machine benchmark on dedicated (non-shared) hardware, including a true cold-network-fetch scenario, to get a trustworthy macro-scale multiplier instead of the noisy 1.01x–1.37x range reported here; (2) collecting real `classifyMs` data from live Gmail/Supabase traffic now that the telemetry exists; (3) investigating length-bucketed batching (sorting sentences/emails by length before batching, to reduce the padding overhead that erased the embedding-batching win and caused the sentiment-batching regression) as a follow-up that might recover a real benefit from batching without the measured downside; (4) generalizing the singleton-

reuse fix into a small reusable "warm model pool" library, since the same pattern is likely to recur as more productivity tools adopt local/hybrid NLP pipelines for privacy reasons, per the motivation in Rojas-Galeano (2024), Kamberi (2026), the Distil Labs model card, and both IEEE baselines (Section 2.1); (5) requesting IEEE Xplore full-text access to verify the quantitative results of La Min Ei & Yongsiriwit (2025) beyond their public abstract/introduction.

References

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8. Xenova / Hugging Face. `distilbert-base-uncased-finetuned-sst-2-english` and `all-MiniLM-L6-v2` (ONNX ports used by `@xenova/transformers`), used as the local models under study in this paper.

> ***On source access:*** *references 1–2 were read directly from their IEEE Xplore pages (title, authors, venue, and either the full public abstract [ref. 1] or the public introduction [ref. 2] — ref. 2's results section is paywalled and its specific accuracy figures are therefore not claimed here). References 3–6 were likewise confirmed against arXiv/thesis-repository pages rather than quoted from search-engine summaries alone.*

Appendix A — Reproducing the benchmarks

Unlike an earlier draft of this paper, the realistic-scale benchmark is a real, checked-in, version-controlled file, not a throwaway snippet reproduced only in this document:

`apps/api/src/scripts/benchmarkClassifier.ts`. It contains, in full, with comments:

- The exact pre-fix `OldEmailClassifier`, recovered verbatim from git history (`git show 50753ab^:apps/api/src/AI_Services/gmail_AI/emailClassifier.ts`), kept only in this file for comparison.
- The synthetic-but-realistic 200-email generator used for Section 6.3/7.2.
- The timing harness (mean/median/stdev/max) used to produce the Section 7.2 table.

Reproduce it with:

```
cd apps/api
npx ts-node --compiler-options '{"module":"commonjs"}'
src/scripts/benchmarkClassifier.ts
# optional: BENCH_EMAILS=200 BENCH_REQUESTS=3 BENCH_TRIALS=5 npx ts-node ... (same
command)
```

The isolated micro-benchmarks in Section 6.1/7.1 (re-initialization cost; uniform- vs. variable-length embedding batching; sentiment batching) were smaller, throwaway scripts written directly against `@xenova/transformers`' `pipeline()` function with no mocks, run outside the repository; their logic is described in Section 6.1 precisely enough to reconstruct, and is intentionally not duplicated as a second checked-in script to avoid maintaining two near-identical harnesses — `benchmarkClassifier.ts` is the one meant to be re-run and extended.

Appendix B — Files changed

- `apps/api/src/AI_Services/gmail_AI/emailClassifier.ts` — singleton reuse; length-bucketed batching for both embedding and sentiment (see Addendum, Section 11); a measured fix to the default semantic-similarity threshold (0.6 → 0.35); a new `filterEmailsWithReasons()` method for explainability and accuracy evaluation, alongside the unchanged `filterEmails()` contract
- `apps/api/src/AI_Services/models.ts` (no content change; now invoked)
- `apps/api/src/server.ts` — warms models at boot
- `apps/api/src/routes/gmail/webhooks/gmailPull.ts` (`classifyMs` telemetry, Section 5.3/7.3)
- `apps/api/src/scripts/benchmarkClassifier.ts` (new) — reproducible realistic-scale latency benchmark, Section 6.3/7.2
- `apps/api/src/scripts/evalSentimentModel.ts` (new) — classification-quality benchmark against the real SST-2 validation set, Section 7.4
- `apps/api/src/scripts/evalFilteringAccuracy.ts` (new) — end-to-end filtering-decision accuracy benchmark against a labeled test set; see Addendum, Section 11
- `apps/api/src/__tests__/emailClassifier.test.ts` (new) — correctness tests for batching, sentiment, and the explainability method

11. Addendum — Findings After Initial Submission

Two further rounds of measurement, after the results above were first written up, changed what's actually shipped. Reported here rather than silently folded into the sections above, so the paper's own

history stays auditable.

Length-bucketed batching (resolves the Section 5.2b revert). The naive full-batch sentiment call described in 5.2b was a measured regression and was reverted to per-email calls. A follow-up experiment tried *length-bucketed* batching instead — sort texts by length, batch within groups of ≤16 — on the theory that naive batching's problem was padding waste from mixing short and long texts in one call, not batching itself. Measured in 3-trial isolated tests with clean, non-overlapping distributions: ~1.14x for sentiment, ~1.21–1.24x for embedding, beating both sequential calls and the naive full batch. This is now shipped (`callInLengthBuckets()` in `emailClassifier.ts`), replacing both the reverted naive-sentiment-batch attempt and the original per-email sentiment calls.

A real filtering-accuracy bug, found and fixed. Neither IEEE baseline, nor any cited work, nor the original Skyler AI code, had ever measured whether the product's actual multi-signal decision (sentiment AND (keyword OR semantic similarity)) correctly keeps/discards a real email — only single-model, single-label accuracy had been measured anywhere in this comparison. A new labeled test set (90 synthetic emails, ground truth assigned by construction across a sentiment × topic grid, not human eyeballing — see `apps/api/src/scripts/evalFilteringAccuracy.ts`) was run against the real, deployed `EmailClassifier`. Result: the shipped default `similarityThreshold` of 0.6 gave 100% precision but only 43.3% recall (F1 60.5%) — verified by direct cosine-similarity measurement that genuinely on-topic paraphrases (no literal keyword) score 0.25–0.54 against single-word keywords, mostly below the 0.6 cutoff, so the semantic-matching path almost never fired in practice and the rule silently degraded to keyword-only matching. A threshold sweep (0.6 → 0.25) found 0.35 as the measured-best point: 94.4% accuracy, 100% precision (no regression), 83.3% recall, 90.9% F1 — a +30.4 F1-point fix, now shipped as the new default.

A new `filterEmailsWithReasons()` method was added alongside the unchanged `filterEmails()` to support this measurement and future explainability UI (per-email sentiment label, matched keyword, and semantic-match boolean), without changing the existing `/gmail/pull` response contract.

A condensed, two-page version of these findings alongside the original results is published separately as `paper/outcomes-summary.pdf`.